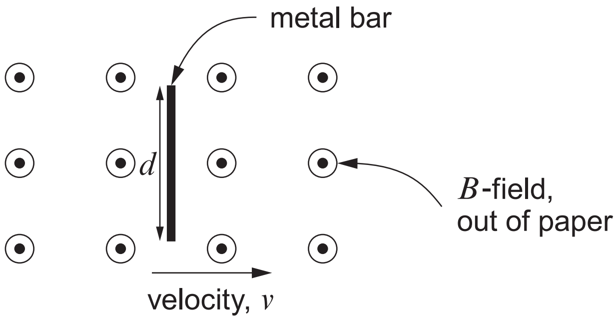
Examiner only

6. (a) The diagram shows a metal bar moving with velocity, v, through a uniform magnetic field, B.



Show that the rate of cutting of flux is Bvd. [2]

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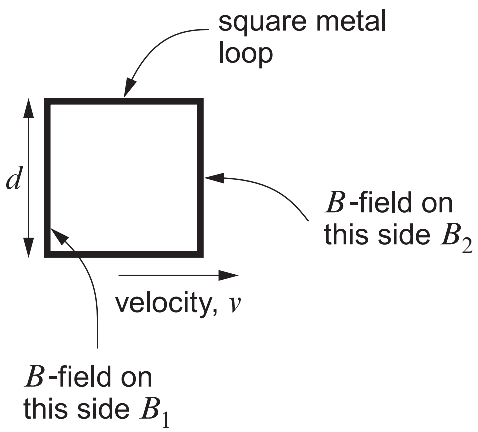
(b) A square metal loop moves through a **non-uniform** B-field with the field coming out of the paper.



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(i) Use the expression in part (a) to explain why the induced emf in the square metallic loop is given by: [3]

$$(B_2 - B_1)vd$$

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(ii) The B-field, through which the square metal loop moves, increases with distance at a rate of 1.20 T m^{-1} (in the direction of the velocity). The square loop has sides of length 3.8 cm and moves at a speed of 8.8 m s^{-1}. Show that the induced emf in the square metal loop is approximately 15 mV. [2]

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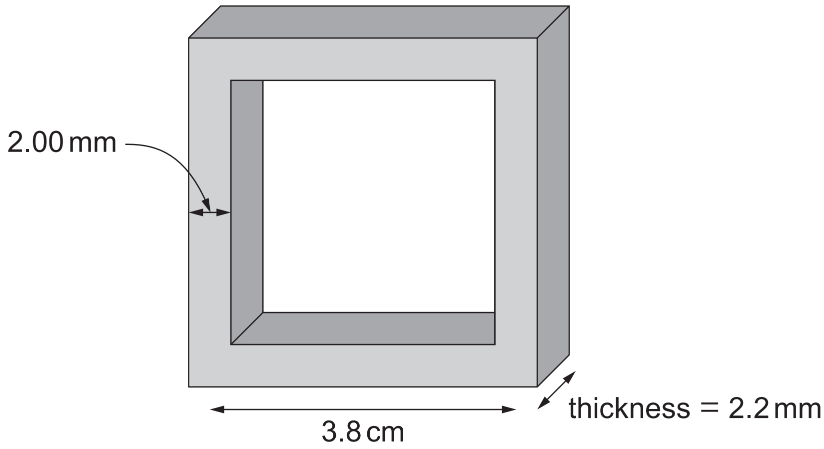
(iii) The square metal loop is made of silver of resistivity $1.59 \times 10^{-8}\,\Omega\,m$ and its dimensions are shown in the diagram below:



Calculate the current in the square loop. [5]

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